

Mohammad Savari

Ph.D. Candidate in Mechanical Engineering
University of Ottawa · Applied Complexity Group

Network Science

Complex Systems

System Dynamics

✉ Moh.Savari@outlook.com 🌐 mohammadsavari.github.io



Scan for full profile

"Exploring how ideas and behaviours cascade through complex networks"

Overview

What this deck covers

1 About Me

2 Experience

3 Education & Skills

4 Research & Publications

5 Key Achievements

6 Contact

01

About Me

Professional Summary

Where engineering meets computational social science

- ▶ **Ph.D. Candidate** in Mechanical Engineering (2021–2026), University of Ottawa — admitted **directly from a bachelor's degree**
- ▶ Researching **social contagion** and **complex networks** with Prof. Roland Bouffanais, *Applied Complexity Group*
- ▶ Second research track at the **Telfer School of Management** with Prof. José Carlos Marques
- ▶ Bridging **network science**, **system dynamics**, and **computational modelling**
- ▶ Recipient of a **full admission scholarship** and a **merit scholarship**

5 yrs

Complex-systems research

3

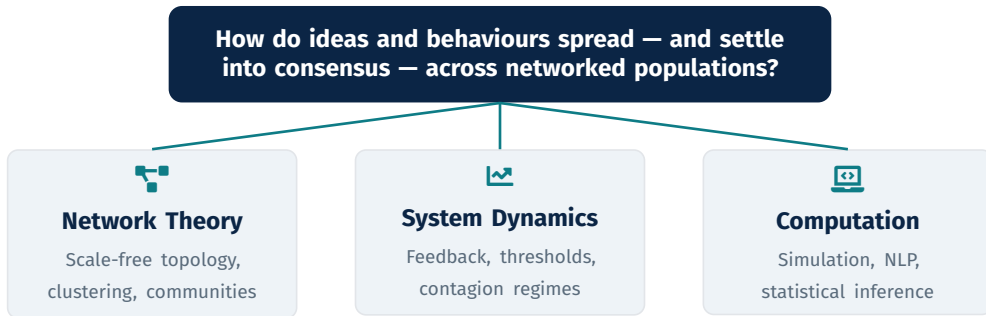
Publications & preprints

6

Courses taught as TA

Research Identity

One question, three lenses



02

Experience

Professional Experience

Industry and engineering-management roles

| Project Manager

Sep 2024 – Present

Centre for Entrepreneurship and Engineering Design (CEED), Ottawa

- ▶ Lead **4 student teams** building accessibility devices, end to end
- ▶ Own procurement, budget tracking, and design-milestone monitoring

| Search Engine Evaluator

Jun 2024 – Present

TELUS International

- ▶ Assess search relevance, accuracy, and content quality at scale
- ▶ Surface behavioural trends that feed back into ranking algorithms

| System Engineering Intern

Sep 2023 – Mar 2024

Airshare Inc., Ottawa

- ▶ Built system models and strengthened cross-team communication frameworks

Academic Experience

Research and teaching

Complex Networks Researcher

May 2025 – Present

Telfer School of Management — with Prof. José Carlos Marques

- ▶ Map the global network of voluntary sustainability standards organisations
- ▶ Web scraping, NLP analysis, and network visualisation in **Python** and **Gephi**
- ▶ Characterise sub-networks, communities, and topological properties

Social Contagion Researcher

Sep 2021 – Present

Faculty of Engineering — Applied Complexity Group

- ▶ Developed a new method for integrating system dynamics into contagion analysis
- ▶ Generated heterogeneous scale-free networks with **tuneable clustering**
- ▶ Performed distribution-likelihood analyses on empirical and synthetic networks

Teaching Assistant

Jan 2022 – Present

University of Ottawa

Mechatronics

System Dynamics

Biomedical System Dynamics

Product Development

ML in Control

Robotics

03

Education & Skills

Education

A direct path into complex systems research

| Ph.D. in Mechanical Engineering

2021 – 2026 (expected)

University of Ottawa, Canada

- ▶ **Thesis:** *Fostering Consensus using Social Contagion*
- ▶ **Direct-Ph.D. admission** from a bachelor's degree — no intervening master's
- ▶ Full admission scholarship **and** merit scholarship

| B.Sc. in Electrical Engineering

2014 – 2018

Control systems, electronics, systems design, and signal processing

Technical Skills

Tools, methods, and the work they support

Programming & Analysis

Python

R

MATLAB

LabVIEW

Statistical Analysis

Gephi

Research Methods

Complex Systems Modelling

Social Contagion Analysis

Network Generation

NLP & Web Scraping

Data Pipelines

Embedded & Hardware

Atmel

Arduino

PLC Platforms

 **Also:** project management · team leadership · technical communication · interdisciplinary collaboration

04

Research & Publications

Research Focus

Thesis: Fostering Consensus using Social Contagion

Core research questions

- ▶ How do social contagions shape **collective consensus** on scale-free networks?
- ▶ Where is the transition from **simple** to **complex** contagion, and what governs it?
- ▶ How does system-dynamics feedback alter information propagation?
- ▶ What mechanisms drive network formation and evolution?

Why it matters

Consensus dynamics underpin everything from public-health messaging and misinformation control to coordinated behaviour in robotic swarms.

Understanding the topology-dependence of contagion tells us *where* to intervene.

Publications

Peer-reviewed articles, preprints, and conference work

2025 How social contagions shape collective consensus in the presence of scale-free networks

M. Savari, R. Bouffanais

Research Square — Preprint · doi:10.21203/rs.3.rs-7635636/v1

2025 Leveraging social contagion to foster consensus in collective decision-making

M. Savari, R. Bouffanais

Int. Conf. on Computational Science (ICCS), Singapore · Conference presentation, Jul 2025

2023 Adaptivity: a path towards general swarm intelligence?

H. L. Kwa, J. L. Kit, N. Horsevad, J. Philippot, M. Savari, R. Bouffanais

Frontiers in Robotics and AI, 10:1163185 · doi:10.3389/frobt.2023.1163185

📖 Full list, abstracts, and PDFs at mohammadsavari.github.io/publications

Selected Projects

Applied research and engineering leadership

VSSO Network Analysis

Mapping the global network of voluntary sustainability standards organisations — who links to whom, which communities form, and how governance authority concentrates.

Telfer School of Management
with Prof. José Carlos Marques

Python

Gephi

NLP

CEED Accessibility Devices

Portfolio of accessibility-focused design initiatives delivered by four student teams, from requirements and procurement through prototype validation.

Centre for Entrepreneurship
and Engineering Design, uOttawa

Project Mgmt

Design

05

Key Achievements

Key Achievements

Highlights across research, academia, and practice

- 🏆 **Direct Ph.D. entry** from a bachelor's degree — awarded for academic excellence
- 🏆 **Full admission & merit scholarships**, University of Ottawa
- 🏆 **Novel methodology** integrating system dynamics into social contagion analysis

- 🏆 **International conference presentation** at ICCS 2025, Singapore
- 🏆 **Co-author** in *Frontiers in Robotics and AI* on swarm adaptivity
- 🏆 **Multi-disciplinary reach** across network science, system dynamics, and engineering management

06

Contact

Get in Touch

Open to research collaborations, academic opportunities, and engineering roles

Mohammad Savari

Ph.D. Candidate, University of Ottawa

✉ Moh.Savari@outlook.com

🌐 mohammadsavari.github.io

🆔    ORCID · Google Scholar · GitHub · LinkedIn — all linked from the site

Interests

Complex Networks

Social Contagion

System Dynamics

Collective Behaviour

Computational Modelling

Data Science



📱 Scan to visit

mohammadsavari.github.io

Thank You

Exploring the dynamics of complex networks

"In complexity lies the key to understanding collective behaviour"



Moh.Savari@outlook.com



mohammadsavari.github.io